

Sagar Marthandan

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Authorized to work in Germany · Immediately available



SUMMARY

Data Analytics Engineer with M.Sc. in Computer Science and Quantitative Economics from CAU Kiel. Experienced building modern data stacks using dbt, SQL, Python, Snowflake, and Apache Airflow. Leverages a systems programming background at IBM in mainframe infrastructure and messaging pipelines to design high-performance, robust, and reliable data systems.

TECHNICAL SKILLS

AI & Machine Learning: Generative AI · RAG (Retrieval-Augmented Generation) · LangChain · LangGraph · Vector Databases (FAISS) · Ollama · Qwen · Prompt Engineering

Data Modeling & Transformation: dbt (Core & Cloud) · Star Schema · Dimensional Modeling · Kimball · SCD Type 2 · Incremental Modeling · ELT/ETL

Programming & Query Languages: Python · SQL · Advanced SQL (CTEs, Window Functions) · PySpark · Spark SQL · DAX · Power Query

Cloud & Data Warehousing: Snowflake · BigQuery (GCP) · AWS (S3, IAM) · PostgreSQL · Azure Databricks

Orchestration & DevOps: Apache Airflow · Dagster · Docker · Terraform · GitHub Actions · Git

Data Quality & Governance: dbt Tests · dbt_expectations · Soda Core · Data Lineage · Data Catalog · RBAC

BI & Visualization: Power BI · DAX · Apache Superset · Looker Studio · Tableau

PROFESSIONAL EXPERIENCE

IBM India Private Ltd

08/2014 - 12/2018

Systems Programmer L2 – CICS Data & Messaging Infrastructure

- Maintained CICS/Db2 transaction infrastructure and IBM MQ messaging pipelines for data flow.
- Introduced CICS audits and MQ monitoring to measure transaction latency and catch issues early.
- Used SMF and MQ event data to support capacity planning and legacy data pipeline modernization.
- Automated CICS/MQ maintenance tasks using SMF insights, cutting platform overhead by 15%.

Staff 4 Cruise, Kiel

10/2019 - 12/2023

Supply Chain Coordinator & Supervisor – Working Student

- Supervised inventory and personnel across 2 AIDA vessels, maintaining zero stockout incidents.
- Coordinated shipments and production plans with internal departments to streamline operations.

PROJECTS

RAG-Based PDF Query Assistant ([[GitHub](#)]) — Engineered a RAG pipeline ingesting PDFs via PyPDF2 into a FAISS vector store via LangChain and OpenAI. Segmented text using character-level splitters to preserve context. Configured local offline Qwen via Ollama to eliminate API cost and guarantee data privacy.

ATS Resume Pipeline Streamlit App ([[GitHub](#)]) — Built a Streamlit application automating resume tailoring and ATS optimization via a 6-stage pipeline. Integrated OpenAI API client for model discovery and context-aware rewriting. Engineered interactive state-machine terminals and automated pdflatex compiling using PowerShell.

NYC Taxi Analytics Pipeline (2025) ([[GitHub](#)]) — Engineered a production-grade ELT pipeline ingesting ~37M NYC Taxi trip records into BigQuery via Airbyte. Modeled dbt mart layers with incremental materialization delivering KPIs. Provisioned GCP infrastructure via Terraform and automated CI/CD via GitHub Actions.

Airbnb Analytics Engineering MDS Pipeline ([[GitHub](#)]) — Built MDS ELT pipeline across staging, core, and mart layers using dbt on Snowflake. Implemented SCD Type 2 snapshots for customer behavior tracking across 3 historical data layers. Cut compute with incremental fact models and validated SLAs via Dagster.

Adventure Works: Global Sales & Returns Dashboard ([[GitHub](#)]) — Designed Star Schema with 1 fact and 7 dimension tables in Power BI. Built KPI-driven DAX measures powering executive sales reports. Delivered scenario planning models using What-If parameters to simulate price adjustment impact.

YouTube E2E: Advanced Data Engineering Pipeline ([GitHub]) — Built production ETL platform: Docker Compose stack with Airflow, PostgreSQL, and Redis. Developed multi-DAG workflows extracting YouTube API v3 analytics. Enforced data quality checks via Soda Core covering schema validation and anomaly detection.

EDUCATION

M.Sc. in Quantitative Economics	<i>Christian-Albrecht-University of Kiel, Germany</i>	<i>present</i>
M.Sc. in Computer Science	<i>Christian-Albrecht-University of Kiel, Germany</i>	<i>11/2023</i>
B.Tech in Information Technology	<i>National Institute of Technology Karnataka, India</i>	<i>05/2014</i>

SPOKEN LANGUAGES

English (C2) · German (B1)